

Exercise on Dictionary

1. Create an empty dictionary with a name D.
2. Create a dictionary using numbers 1 to 5.
 - (a) key:value pairs can be integers. Print.
 - (b) keys as strings and value as integers. Print.
 - (c) Create a dictionary as in 2(b). Print it. Then, change its values to a new value by adding 1 with existing values by accessing each key. Print new dictionary.
3. Create a dictionary with your profile details like name, age, date of birth, gender, qualification and home town as keys and values as applicable. Print it. Add a new key "Occupation" with its value and Print it.
4. Create a dictionary as in question no. 3 and do the following.
 - (a) Print the value associated with each key using the get() function.
 - (b) Repeat 4(a), but store the value returned by get() in a variable and print it.
 - (c) Print a value associated with the key "Marital status". Note what happens?. Write down the result shown.
 - (d) If "None" was the print value for (c), Add a value "No information" for the key "Marital status" and print the dictionary to see whether it is added. Do not use direct assignment (d[key] = value). Use methods like setdefault() or update().
 - (e) Print all the keys associated with the dictionary in a list format.
 - (f) Print all the values associated with the dictionary in a list format.
 - (g) Print "Key-value" pairs as a list of tuple.

(h) Print the keys, values and key:value pairs separately using iteration.

(i) Create a dictionary as in question no. 3. Update the dictionary with details such as Father's name, Mother's name, Income, favourite subject.

(j) Create a dictionary as in question no. 3, update as in 4(i). Remove the items recently Updated i.e., Father's name, mother's name, Income, favourite subject. Items to be removed has to be given as a list. Using iterations remove the items. After the loop is executed, print the removed items in a list.

(k) Do the same exercise as in 4(j). Construct a dictionary called removed_items with Removed key-value pairs.

5. Create a dictionary with 4 names of your choice and their ID number and age.

(i) Sort the names based on the alphabetical order and print using items() and for Loop.

(ii) Check whether the given name exists in the dictionary using "in" .

6. Create a dictionary with roll numbers from 1 to 6 as keys and student names as values. Include your name and your friends' names.

Accept a roll number randomly (or from user input) and check whether the roll number exists in the dictionary.

- If the roll number exists, print the corresponding student name.
- Otherwise, print the statement: "The given roll number does not exist".

7. Create a dictionary using fromkeys() with "a", "b", "c" as keys and values

Follows.

(a) value as "11"

(b) value as "[1, 2, 3]"

- (c) value as [] (empty list)
- (d) After creating an empty list as value change the value associated with "b" by using append(1). See what happens.

8. Create a dictionary using dictionary comprehension.

- (a) integers 1 to 4 as keys and values the same as keys.
- (b) integers 1 to 4 as keys and its squares as values.
- (c) use each letter in the list ["a", "b", "c"] for key and empty list as value
- (d) Repeat 8(c) and modify value associated with "b" by using append(1).
- (e) compare 7(d) and 8(d) and think why if you find a difference.

9. Frequency of occurrence can be found using dictionary. Let us do it following the Steps below.

Let D = ["a", "b", "c", "b", "d", "a", "d", "d", "c"]

We will find how many times each alphabet in the list "D" occurred.

Hints are below.

- (a) We will have our counting of occurrence as values corresponding to each Unique alphabet in the list "D" as keys. Initially create an empty dictionary with name of your choice.
- (b) Iterate over each letter in the list "D"
- (c) With the above step we have got our keys. Check if the letter exists in the final dictionary created in step(a). if exists increase value by 1 otherwise assign the value 1.
- (d) Print the dictionary created in step a.
- (e) Do the exercise using get() method [Hint. Instead of step C use get()]

10. Create a nested dictionary called course containing details of two students.

For each student, include the following keys:

- name
- roll_number

- studies
- subjects
- marks

Requirements:

1. The value of the key studies must itself be a dictionary containing:
 - 10th
 - 12th
 - UGas keys, and the corresponding school/college names as values.
2. The value of the key subjects must be a list containing the subjects studied in 12th standard.
3. The value of the key marks must be a dictionary where:
 - subject names are keys
 - marks obtained are values
4. Print the complete dictionary.
5. Access and print:
 - the UG college name of the second student
 - marks scored in a particular subject of your choice of the first student
 - all subjects studied by a student of your choice